Dimension Reduction

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Need of dimension reduction

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Why is dimension reduction needed?

- In real world problems we may get data with very high dimension which makes it difficult to store , visualize and use for machine learning model (E.g. similar outfit visualization, medical data visualization

- Data might consist of features which provide similar information e.g. correlated features

- Data might have redundant features e.g. feature which provides height of person in inches and another feature which provides height in the cm

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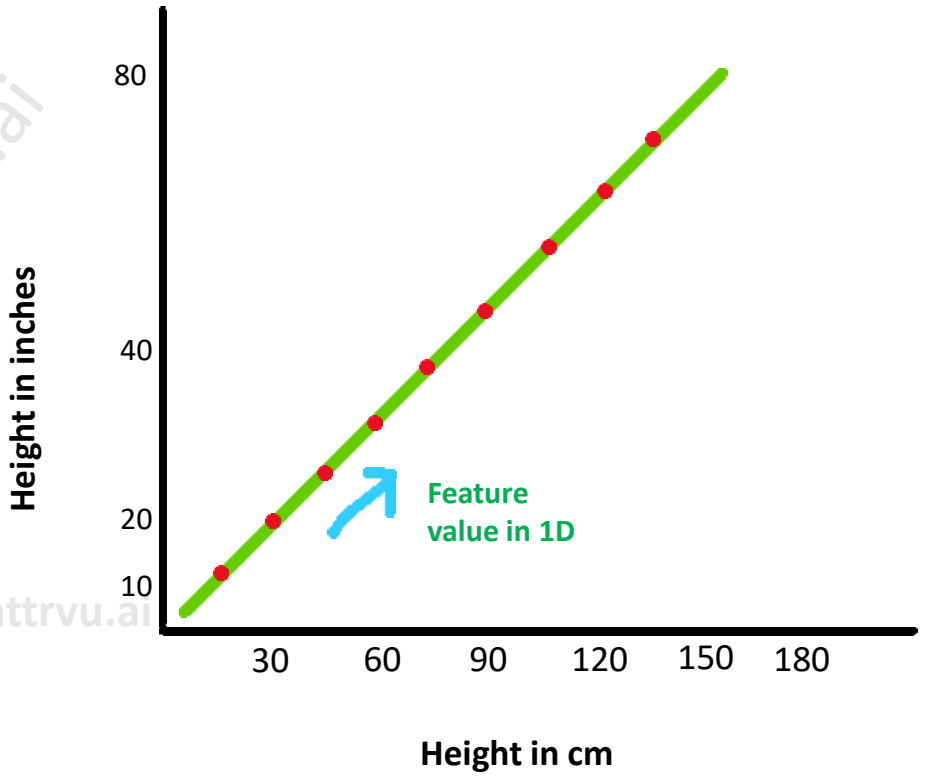
Dimension reduction

- X and Y axis same feature – height of person but measures in cms and inches respectively
- Both the features are not needed, and we can use any one of them
- Using dimension reduction both the features can be converted from 2D to 1D by taking location on green line as the feature value



In real world datasets presence of such redundant features in dataset might not be easy to spot

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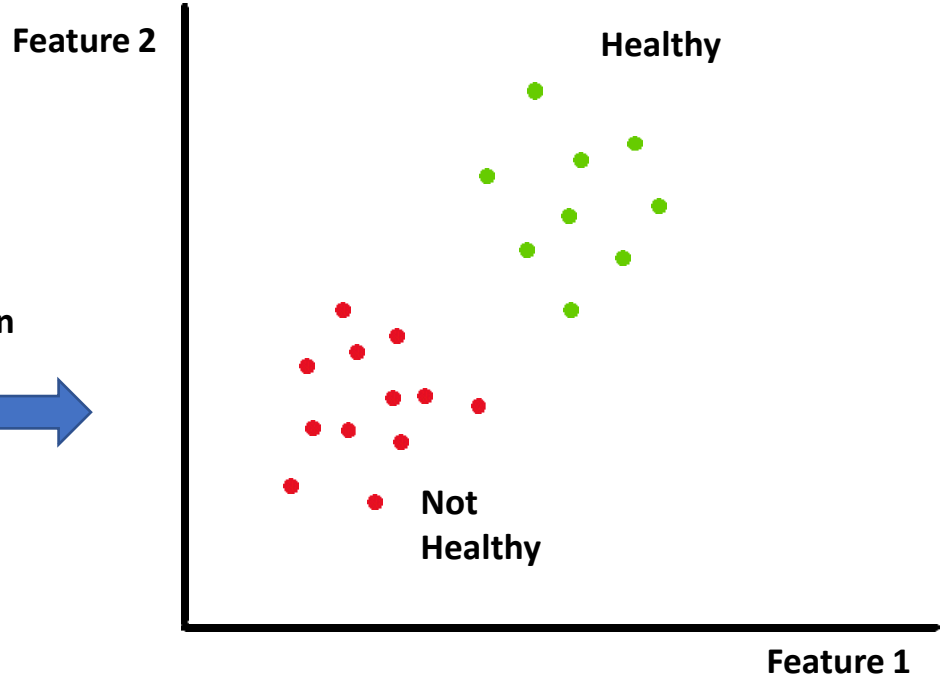
Dimension Reduction

Input data consists of multiple features

Dimension reduction is needed to visualize data in 2D

Persona Id	Height (cm)	Height (inch)	weight	BMI	Daily walking (km)
1	150	60	60	26.70	2.5
2	175	70	65	21.22	3.0
3	200	80	90	27.78	2.2

Dimension reduction



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Principal Component Analysis

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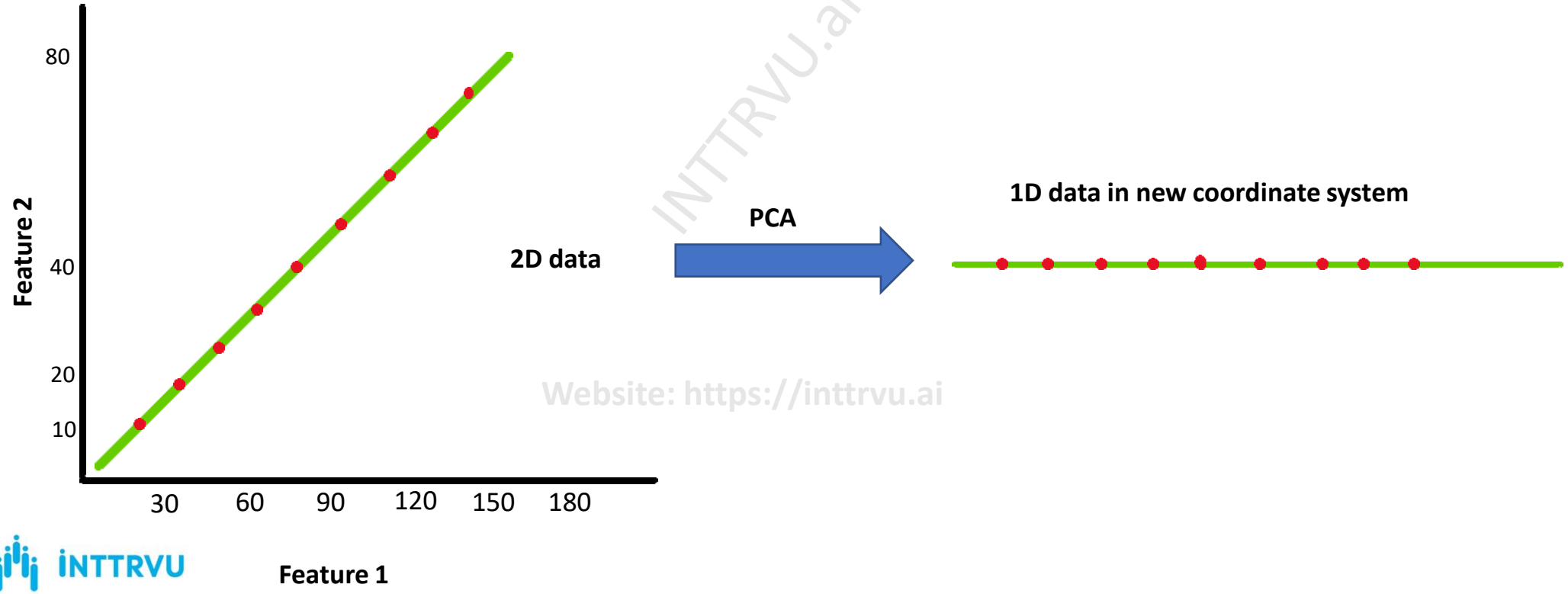
Principal Component Analysis

- Principal Component Analysis (PCA) is unsupervised learning method for dimension reduction
- PCA reduces dimension by linearly transforming the data into a new coordinate system where (most of) the variation in the data can be described with fewer dimensions than the initial data.



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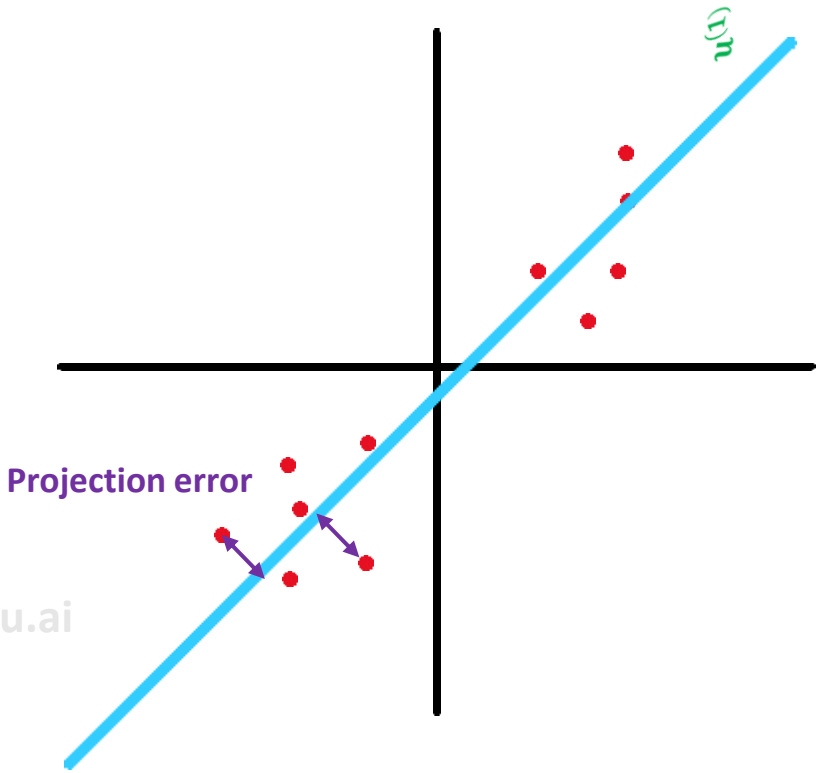
PCA

- From 2-dimesional to 1-dimensional :Find the direction (vector $u^{(1)}$) on which data can be projected to minimize projection error

- From N-dimensional to K-dimensional : Find the directions (k vectors $u^{(1)}$, $u^{(2)}$, …, $u^{(k)}$) on which data can be projected to minimize projection error

PCA is Not a linear regression



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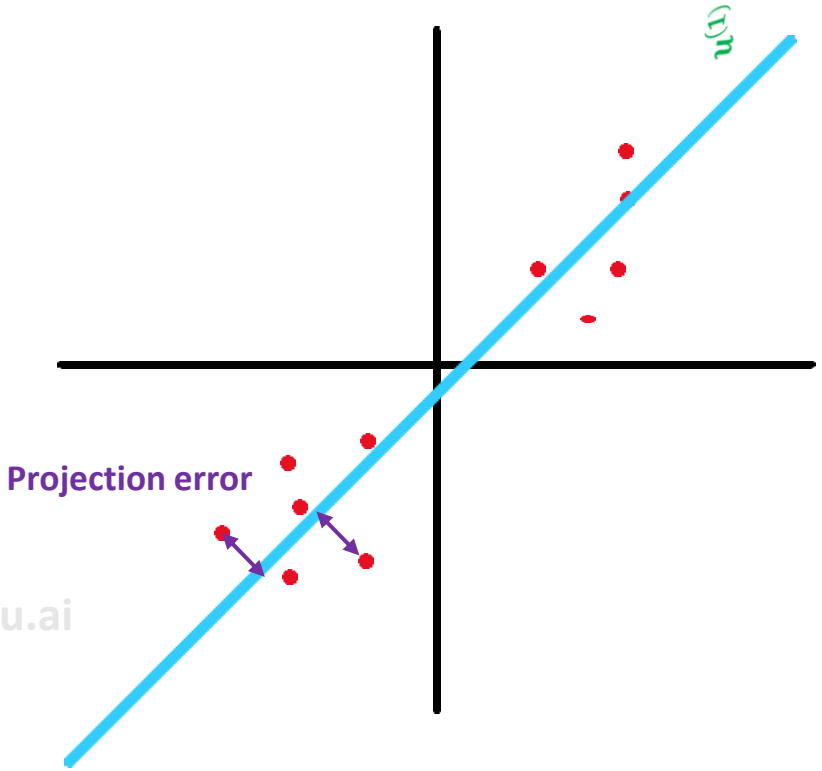
PCA

From N-dimensional to K-dimensional : Find the directions (k vectors $u^{(1)}$, $u^{(2)}$, ..., $u^{(k)}$) on which data can be projected such that:

- Projection error is minimized
- New features are created by projecting to new-coordinate system using linear transformation
- All the vectors are orthogonal (perpendicular) each other
- All the features created are non-correlated



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PCA Steps

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PCA Steps

1) Data pre-processing:

If training set consists of n features $x^{(1)}, x^{(2)}, \ldots, x^{(n)}$

Calculate mean for every feature

$$\mu_i = \frac{1}{m}\sum_{j=1}^{m} x_j^{(i)}$$

- Substitute each feature value $x_j^{(i)}$ by $(x_j^{(i)} - \mu_i)$ (mean normalization)

- If features are different scale – scale the features so that all the features are in similar range

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PCA Steps

2) Calculate covariance matrix:

Covariance is a measure of the relationship between two random variables and to what extent, they change together

$$\Sigma = \frac{1}{n-1}\left((X-\mu)\cdot(X-\mu)^T\right)$$

n is number of features

X is matrix of input feature vectors (original data)

μ is matrix of mean values of features (original data)

Covariance is a measure of the relationship between two random variables and to what extent, they change

$$\begin{bmatrix} Var(x_1) & \dots & Cov(x_1, x_n) \\ \vdots & \ddots & \vdots \\ Cov(x_n, x_1) & \dots & Var(x_n) \end{bmatrix}.$$

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Eigen values and Eigen vectors: Refresher for PCA

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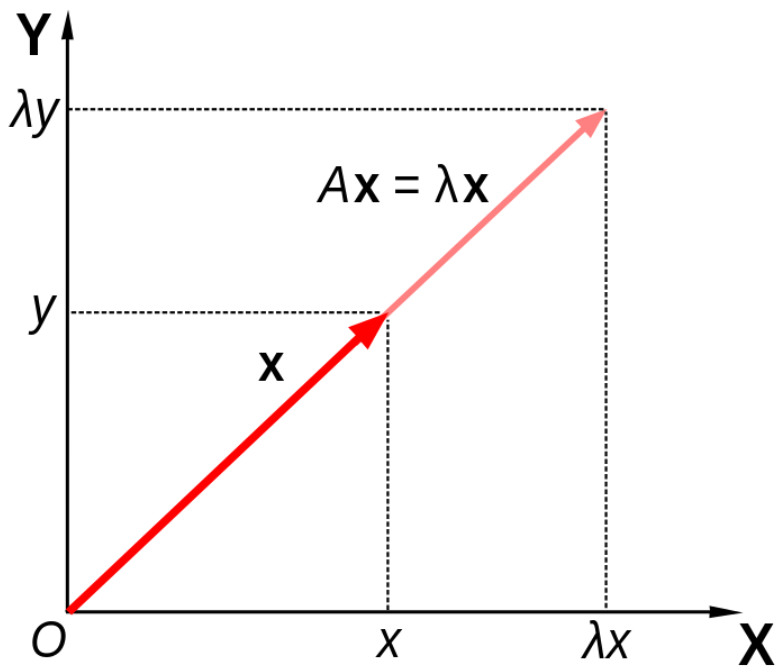
Eigen Value and Eigen Vectors

Eigenvectors are the vectors (non-zero) that do not change the direction when any linear transformation is applied

Any linear transformation applied to eigen vector results in only change in magnitude but not the direction

Change in magnitude is represented by a scaler value called as eigen value



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Eigen Value and Eigen Vectors

Suppose $\boldsymbol{x}$ is a vector in vector space V

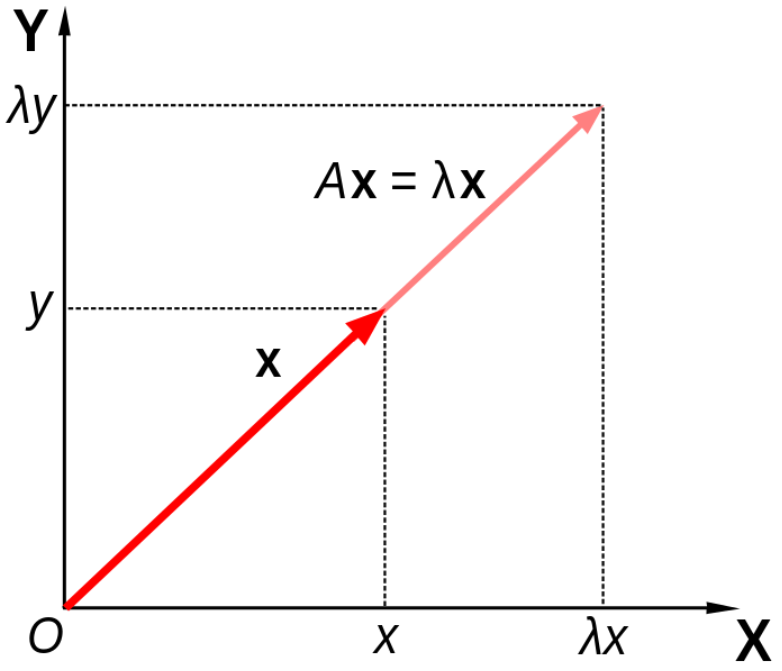
A is linear transformation applied to $\boldsymbol{x}$ (A is n*n matrix)

Then if :

$$A\boldsymbol{x} = \lambda\boldsymbol{x}$$

$\boldsymbol{x}$ is called as eigen vector of A (x is n*1 vector/matrix)

λ is a scalar and called as eigen value of A



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PCA Continued

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PCA Steps

3) Eigen value decomposition

- Calculate Eigen values and Eigen vectors of the covariance matrix
- Eigen vectors represent orthogonal vectors on which data can be projected
- Eigen values represent amount of variance explained by the corresponding eigen vector

Variance explained by each vector k = Eigen value of vector k / sum of all eigen values

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PCA Steps

4) Select k (number of component /dimension of new data)

- Select K which explains at least 90% variance of the data
- Higher the explained variance – lower the projection error

E.g. If PCA is applied to data with 100 features and sum of first 40 eigen values gives more than 90% variance then we can select first 40 eigen vectors (dimensions)

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Tips for applying PCA

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Useful tips for applying PCA

- Never use PCA just to reduce the dimension if original features can be handed by the model
- Use of PCA reduces model explainability as we don't know which features have contributed to the creation of each principal component
- Select number of components based on explained variance and number of features which model can handle – never select using elbow curve
- Apply PCA only on training data and use the learned transformations on testing and prediction data. Never apply PCA on whole data and then do train/test split

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Questions?

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